

i Forside

UNIVERSITY OF OSLO

Faculty of mathematics and natural sciences

Mid term exam IN1900, IN-KJM1900, IN-GEO1900

Date: Oktober 10, 2025, 15:00-19:00

Attachments: None

Permitted aids: None. A calculator is available in Inspira.

- The exam set consists of 20 multiple choice questions. The maximum score is 25.
- All code in the exam is written in Python 3.
- Most of the questions have one correct answer. Correct answer gives 1 point, wrong answer gives 0 points.
- Questions 8, 10, 11, 13 and 20 have 4 answers each. Each correct answer gives 0.5 points, and 0 points is given for a wrong answer or if no answer is given. Maximum 2 points for each of the questions.
- There are no negative points for wrong answers, so all questions should be answered.

1 Hva skrives ut?

What is printed in the terminal when this code is run?

```
teller = 0
while teller < 4:
    teller = teller + 1

print(teller)
```

Select one alternative:

- ☐ 3
- ☐ En feilmelding
- ☐ 4
- ☐ 5



Maximum marks: 1

2 Hva skrives ut?

What is printed in the terminal when this code is run?

```
a = list(range(2, 11, 3))  
print(a)
```

Select one alternative:

☐ [2, 4, 6, 8, 10]

☐ [2, 11, 3]

☐ [2, 5, 8]



☐ [2, 5, 11]

Maximum marks: 1

3 Hva skrives ut?

What is printed when this code is run?

```
data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]  
print(data[1][2])
```

Select one alternative:

☐ 5

☐ 6



☐ [4, 5, 6]

☐ 2

Maximum marks: 1

4 Hva skrives ut?

What is printed in the terminal window when the following code is run?

```
a = 5  
b = a * 2  
a = a + b  
print(a, b)
```

Select one alternative:

☐ 15 10



☐ 15 30

☐ 10 10

☐ An error message

Maximum marks: 1

5 Hva skrives ut?

What is printed in the terminal when this code is run?

```
C = -20  
dC = 10
```

```
Celsius = []  
Fahrenheit = []
```

```
while C <= 40:  
    Celsius.append(C)  
    Fahrenheit.append(2 * C + 30)  
    C = C + dC
```

```
print(Celsius[-1], Fahrenheit[-1])
```

Select one alternative:

- ☐ 45 120
- ☐ 40 110
- ☐ An error message
- ☐ 35 100



Maximum marks: 1

6 Hva skrives ut?

What is printed when this code is run?

```
line = 'All work and no play makes Jack a dull boy.'  
words = line.split()  
print(f'{words[0]} {words[-2]}')
```

Select one alternative:

- ☐ An error message
- ☐ A y
- ☐ All work dull boy
- ☐ All dull



Maximum marks: 1

7 Hva skrives ut?

What is printed when this code is run?

```
labels = ['a', 'b', 'c']
```

```
for i in range(3):  
    print(labels[i], end = ' ')  
    for j in range(i, 3):  
        print(j, end = ' ')
```

The argument `end = ' '` to the print function makes each print end with space (' ') instead of a new line.

Select one alternative:

☐ a 0 1 2 b 1 2 c 2



☐ a 1 2 3 b 2 3 c 3

☐ 0 0 1 2 1 1 2 2 2

☐ a 0 1 b 0 1 2 c 0 1 2 3

Maximum marks: 1

8 Boolske uttrykk

What are the values of the boolean expressions?

Please match the values:

	False	True
4 >= 3 and not True	☐ ✓	☐
True and False or True	☐	☐ ✓
True or False	☐	☐ ✓
True and False or False	☐ ✓	☐

Maximum marks: 2

9 Hva skrives ut?

What is printed when this code is run?

```
from math import sqrt  
x = 10
```

```
def abc(a,b,c):  
    x = (-b - sqrt(b**2 - 4 * a * c))/(2 * a)  
    y = (-b + sqrt(b**2 - 4 * a * c))/(2 * a)  
    return x, y
```

```
roots = abc(1, 2, 1)  
print(x)
```

Select one alternative:

- ☐ -1
- ☐ 1
- ☐ (-1, -1)
- ☐ 10



Maximum marks: 1

10 Indeksering i lister

We have the following three lists:

- `kort = [1, 0, -1]`
- `lang = [1, 2, 3, 4, 5]`
- `liste = [1, 2, [1], [2], [[1]], [[2]]]`

To the left of the table are four expressions, and at the top of the table are six possible results when the expressions are evaluated. Choose the right alternative for each expression.

Please match the values:

	1	2	[1]	[2]	[[1]]	[[2]]
<code>liste[-2]</code>	☐	☐	☐	☐	☒	☐
<code>liste[kort[1]]</code>	☒	☐	☐	☐	☐	☐
<code>liste[kort[2] * lang[2]]</code>	☐	☐	☐	☒	☐	☐
<code>liste[lang[kort[0]]]</code>	☐	☐	☒	☐	☐	☐

Maximum marks: 2

11 Lister og Numpy arrays

Assume that this code is run.

```
import numpy as np
q = [x * x for x in range(51)]
p = np.linspace(0, 50, 51)
s = p**2
```

What are the values of the boolean expressions?

Please match the values:

	True	False
<code>s[-1] > 200</code>	☐ ✓	☐
<code>abs(q[-1] - s[-1]) < 0.01</code>	☐ ✓	☐
<code>len(q) == len(s)</code>	☐ ✓	☐
<code>len(p) == 50</code>	☐	☐ ✓

Maximum marks: 2

12 Logistisk vekst

The code below solves a difference equation that describes exponential growth.

```
r = 5.0 # growth rate
N = 20 # number of years
B = np.zeros(N+1)
B[0] = 100
for n in range(1,N+1):
    B[n] = B[n - 1] + (r / 100) * B[n - 1]
```

We want to change the code to solve a logistic model, where the growth approaches zero and the population curve becomes flat when B approaches the carrying capacity R. To achieve this we need to define a new variable R, for instance R = 1000, and change the line inside the for loop. Which of the following lines should go inside the for loop?

Select one alternative:

- ☐ $B[n] = B[n-1] + (r / 100) * B[n-1] - (1 - B[n-1] / R)$
- ☐ $B[n] = B[n-1] - (r / 100) * B[n-1]$
- ☐ $B[n] = B[n-1] + (r / 100) * B[n-1] + (1 - B[n-1] / R)$
- ☒ $B[n] = B[n-1] + (r / 100) * B[n-1] * (1 - B[n-1] / R)$



Maximum marks: 1

13 Lesing fra fil

The file "marathons.txt" contains a collection of marathon times from the period 1981-2022:

2:05:42; Khalid Khannouchi; 1999; Chicago
 2:05:38; Khalid Khannouchi; 2002; London
 2:04:55; Paul Tergat; 2003; Berlin
 2:07:12; Carlos Lopes; 1985; Rotterdam
 2:04:26; Haile Gebrselassie; 2007; Berlin
 2:03:59; Haile Gebrselassie; 2008; Berlin
 2:06:50; Belayneh Dinsamo; 1988; Rotterdam
 2:06:05; Ronaldo da Costa; 1998; Berlin
 2:03:23; Wilson Kipsang; 2013; Berlin
 2:02:57; Dennis Kimetto; 2014; Berlin
 2:08:18; Robert De Castella; 1981; Fukuoka
 2:01:39; Eliud Kipchoge; 2018; Berlin
 2:01:09; Eliud Kipchoge; 2022; Berlin
 2:08:05; Steve Jones; 1984; Chicago
 2:03:38; Patrick Makau; 2011; Berlin

We want to write code that reads such a file and writes out the fastest time and the name of the person achieving it. The output from the code shall look as follows:

2:01:09 Eliud Kipchoge

Two of the codes below are correct and will give this output. Which ones?

Alternative A:

```
fastest = (14400,'4:00:00')
name = 'No record found'
with open('marathons.txt') as infile:
    for line in infile:
        w = line.split(';')
        h,m,s = [float(ti) for ti in w[0].split(':')]
        seconds = h*3600 + m*60 + s
        if seconds < fastest[0]:
            fastest = (seconds,w[0])
            name = w[1]
print(fastest[1], name)
```

Alternative B:

```
fastest = 100000
name = 'No record found'
with open('marathons.txt') as infile:
    for line in infile:
        w = line.split(';')
        h,m,s = [float(ti) for ti in w[0].split(':')]
        seconds = h*3600 + m*60 + s
        if seconds < fastest:
            fastest = seconds
            name = w[1]
print(fastest, name)
```

Alternative C:

```

fastest = (14400, '4:00:00')
name = 'No record found'
with open('marathons.txt') as infile:
    for line in infile:
        w = line.split()
        h,m,s = [float(ti) for ti in w[0].split(':')]
        seconds = h*3600 + m*60 + s
        if seconds < fastest[0]:
            fastest = (seconds,w[0])
            name = w[1]
print(fastest[1], name)

```

Alternative D:

```

fastest = 14400
name = 'No record found'
with open('marathons.txt') as infile:
    for line in infile:
        w = line.split(';')
        h,m,s = [float(ti) for ti in w[0].split(':')]
        seconds = h*3600 + m*60 + s
        if seconds < fastest:
            fastest = seconds
            record = w
print(record[0],record[1])

```

Please match the values:

	Correct	Wrong
Alternative A	☐ ☒	☐
Alternative C	☐	☐ ☒
Alternative D	☐ ☒	☐
Alternative B	☐	☐ ☒

Maximum marks: 2

14 Feilhåndtering med try-except

What is printed when this code is run?

```
A = [['-1','0','1'], ['0','0','0'], ['10','9','8']]
```

```
try:
```

```
    b = int(A[2])
```

```
except IndexError:
```

```
    print(f'A has length {len(A)}')
```

```
    exit()
```

```
except TypeError:
```

```
    print(f'Cannot convert {A[2]} to int')
```

```
    exit()
```

```
print(b)
```

Select one alternative:

- ☒ Cannot convert ['10','9','8'] to int
- ☐ A has length 3
- ☐ 10
- ☐ Cannot convert ['0','0','0'] to int



Maximum marks: 1

15 Hvilken linje mangler?

The file "EUR.txt" contains historical exchange rates and has the following structure:

```
26.09.2025 EUR 11.7105
25.09.2025 EUR 11.6890
24.09.2025 EUR 11.6880
23.09.2025 EUR 11.6720
22.09.2025 EUR 11.7025
19.09.2025 EUR 11.6705
18.09.2025 EUR 11.5870
17.09.2025 EUR 11.6260
```

We want the code below to read such a file and store the exchange rates in a dictionary, with the dates as keys and the corresponding exchange rates as values. One line is missing in the code.

```
rates = {}
```

```
with open('EUR.txt') as infile:
```

```
    for line in infile:
```

```
        words = line.split()
```

```
        #missing line goes here
```

Which line must be added for the code to work as intended?

Select one alternative:

- ☐ rates = {words[0], float(words[-1])}
- ☐ rates.append(float(words[-1]))
- ☒ rates[words[0]] = float(words[-1])
- ☐ rates.append(words[0], words[-1])



Maximum marks: 1

16 Hvilken linje mangler?

Which line needs to be insterted for the test to pass?

```
def triangle_area(v):
    #missing line goes here
    return 0.5*abs(x2*y3-x3*y2-x1*y3+x3*y1+x1*y2+x2*y1)

def test_triangle_area():
    # Verify the area of a triangle with vertices
    # (0,0), (1,0), and (0,2)
    vertices = [(0,0), (1,0), (0,2)]
    expected = 1
    computed = triangle_area(vertices)
    tol = 1E-14
    success = abs(expected - computed) < tol
    msg = f"computed area={computed} != {expected}(expected)"
    assert success, msg
```

test_triangle_area()

You can assume that the expression in the return statement is correct, so that the test will pass if you add the missing line. Which line is missing?

Select one alternative:

- ☐ x1, x2, x3 = v[:3]; y1, y2, y3 = v[3:]
- ☒ x1, y1 = v[0]; x2, y2 = v[1]; x3,y3 = v[2] ✓
- ☐ x = v[0:3]; y = v[3:]
- ☐ x1, y1 = vertices[0]; x2, y2 = vertices[1]; x3,y3 = vertices[2]

Maximum marks: 1

17 Hvilket funksjonskall?

The function **secant(f, x0, x1, tol = 1e-5)** applies the secant method to solve an equation on the form $f(x) = 0$. The parameters to the function are the Python function **f**, which implements the mathematical function $f(x)$, two start values **x0** and **x1**, and a tolerance **tol**.

```
def secant(f, x0, x1, tol = 1e-5):
    while abs(f(x1)) > tol:
        x2 = x1 - f(x1) * (x1 - x0) / (f(x1) - f(x0))
        x0 = x1
        x1 = x2

    return x2, f(x2)
```

We want to use the function to solve the equation $x^2 - 2 = 0$. Which function call is correct?

Select one alternative:

- ☒ x, f = secant(lambda x: x**2-2, 2.0, 1.8) ✓
- ☐ x, f = secant(f(x) = x**2-2, 2.0, 1.8)
- ☐ x, f = secant(f = x**2-2, 2.0, 1.8)
- ☐ x, f = secant(x**2-2, 2.0, 1.8)

Maximum marks: 1

18 Hvilken påstand er riktig?

The two functions below both compute a sum. One of the functions uses a while-loop and the other uses a for-loop.

```
def sum_while(x,n):
```

```
    s = 0; i = 1
```

```
    while i <= n+1:
```

```
        s += x**i/i
```

```
        i += 1
```

```
    return s
```

```
def sum_for(x,n):
```

```
    s = 0
```

```
    for i in range(1,n+1):
```

```
        s += x**i/i
```

```
    return s
```

We call the functions in the following way:

```
s1 = sum_while(5.0, 7)
```

```
s2 = sum_for(5.0, 7)
```

Which statement is correct?

Select one alternative:

- ☐ The while-loop contains an error and will give an error message
- ☐ The two functions will return the same value
- ☐ The for-loop contains an error and will give an error message
- ☐ The while-loop includes more terms than the for-loop



Maximum marks: 1

19 Hva gjør funksjonen?

What does the function **my_fun** do? You can assume that the argument **numbers** is a list of numbers.

```
def my_fun(numbers):  
    n = len(numbers)  
    for i in range(n):  
        for j in range(n-1):  
            if numbers[j] < numbers[j+1]:  
                tmp = numbers[j]  
                numbers[j] = numbers[j+1]  
                numbers[j+1] = tmp  
    return numbers
```

Select one alternative:

- ☐ Returns the same list of numbers, with the same order as the input
- ☒ Returns a list with the numbers sorted in descending order
- ☐ Returns a list with the numbers sorted in ascending order
- ☐ Returns a list with the same numbers, sorted in random order



Maximum marks: 1

20 Hvilke funksjoner er riktige?

We want to implement a function that takes a list as input and returns the largest element in the list. The elements in the list are numbers, which can be positive or negative, and we want the function to support the following use:

```
>>> print(my_max([1,4,5,-1]))
5
>>> print(my_max([-1,-2,-5]))
-1
```

Which of the following implementations of the function **my_max(a)** are correct?

Alternativ A:

```
def my_max(a):
    max_found = a[0]
    for i in range(1,len(a)):
        if a[i] > max_found:
            max_found = a[i]
    return max_found
```

Alternativ B:

```
def my_max(a):
    max_found = 0
    for e in a:
        if e > max_found:
            max_found = e
    return max_found
```

Alternativ C:

```
def my_max(a):
    for i in range(1,len(a)):
        if a[i] > a[i-1]:
            max_found = a[i]
        else:
            max_found = a[i-1]
    return max_found
```

Alternativ D:

```
def my_max(a):
    max_found = a[0]
    i = 0
    while i < len(a):
        if a[i] >= max_found:
            max_found = a[i]
        i += 1
    return max_found
```

Please match the values:

	Wrong	Correct
Alternativ D	☐	☐ ✓
Alternativ B	☐ ✓	☐
Alternativ A	☐	☐ ✓
Alternativ C	☐ ✓	☐

Maximum marks: 2